

JEE MAIN REVISION MASTER: CHEMICAL BONDING AND MOLECULAR STRUCTURE 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 26 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Chemical Bonding And Molecular Structure - Core Application	26 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Chemical Bonding And Molecular Structure - Core Application (Very High Yield — 26 PYQs)

Core Principle: Key problem pattern in Kcet Chemical Bonding And Molecular Structure.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "With respect to resonance structures of CO_3^{2-} ion, which of the following statements are correct? (a) All C-O..." [Ans: N/A]
- [2026 Shift] Q#2 (MCQ): "In which of the following option/options, the order of arrangement does not agree with the variation of property indicated against it? (a) \$..." [Ans: N/A]
- [2026 Shift] Q#3 (MCQ): "The types of hybrid orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ respectively are..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

With respect to resonance structures of CO_3^{2-} ion, which of the following statements are correct? (a) All C-O bonds in CO_3^{2-} are equivalent (b) There are three resonance structures possible for CO_3^{2-} ion (c) The position of carbon and oxygen should change in every resonance structure (d) The formal charge on carbon atom is -2

- (A) a, b and c
- (B) a and b only
- (C) b and d only
- (D) a, b and d

Correct Answer: None

Solution Analysis: In the carbonate ion, CO_3^{2-} , carbon is at the centre and the three oxygen atoms are attached to it. The ion has three possible resonance structures. In each structure, one C=O double bond and two C-O⁻ single bonds are shown. The double bond can be placed with any one of the three oxygen atoms. So statement (b) is correct. In resonance, the actual structure is not any one of these structures. The actual carbonate ion is a resonance hybrid. Because of this, all three C-O bonds become equivalent. Each C-O bond has